

Indian Institute of Engineering Science and Technology, Shibpur
Dual Degree (B. Tech.- M. Tech.) 3rd Semester Final Examination,
November 2015

Subject: Mathematics III (MA-301)

Time: 3 hours

Full Marks: 70

Use a separate answerscript for each half

1st Half (Full Marks - 40)

Answer Question No. 1 and ANY TWO of the rest

1. Answer ANY THREE questions:

$3 \times 6 = 18$

- (a) A fair six-faced die is tossed until all the six faces are observed. If X is the number of tosses required, find $E(X)$.
 - (b) X and Y are stochastically independent random variables each having $N(0,1)$ distribution. What is the distribution of the random variable $U = \frac{X+Y}{X-Y}$? Justify your answer.
 - (c) Define Mean Square Error (MSE) of an estimator. Establish the bias-variance decomposition formula for MSE.
 - (d) Prove that the correlation coefficient between two random variables always lies between -1 and 1.
 - (e) A straight line of unit length is divided into three line segments by randomly choosing two points on it. Find the probability that the three line segments form a triangle.
2. (a) State and prove Tchebycheff's Inequality.
 - (b) Show by Tchebycheff's Inequality that in 2000 throws with a coin, the probability that the number of heads lies between 900 and 1100 is at least 19/20.

$5 + 6 = 11$

3. (a) X, Y are independent random variables having common density

$$f_{\alpha,\beta}(x) = \begin{cases} \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, & \text{for } x > 0 \quad (\alpha, \beta > 0) \\ 0 & \text{otherwise.} \end{cases}$$

Find the probability density function (p.d.f.) of $X + Y$.

- (b) Let T_1 and T_2 be two statistics with $E(T_1) = \theta_1 + \theta_2$ and $E(T_2) = \theta_1 - \theta_2$, find unbiased estimators of the parameters θ_1 and θ_2 .

$6 + 5 = 11$

4. (a) The following is a random sample of size 10 from a uniform distribution on $(0, \theta)$, $\theta > 0$: 5.6, 3.9, 3.2, 4.8, 7.8, 6.1, 10.2, 8.4, 7.1, 9.8. Find the Maximum Likelihood Estimator (M.L.E.) of θ . Justify your answer.
- (b) $X_1, X_2, \dots, X_n$ is a random sample of size n from a distribution having finite variance σ^2 . Prove that $T_n(X_1, X_2, \dots, X_n) = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ is consistent but not unbiased for σ^2 .

$$5 + 6 = 11$$

5. (a) Suppose $f(x)$ and $g(y)$ are two probability density functions. Prove that $h(x, y) = f(x)g(y)$ is a bivariate density with $f(x)$ and $g(y)$ as its marginal densities. Further, show that the two random variables are stochastically independent.
- (b) Derive the normal equations for the simple linear regression model :

$$y_i = \alpha + \beta x_i + e_i, i = 1, 2, \dots, n$$

$$5 + 6 = 11$$

2nd Half (Full Marks - 30)

Answer ANY THREE Questions

6. (a) Find the Laplace transforms of

$$(i) f(t) = \begin{cases} \sin t, & 0 < t < \pi \\ 0, & t > \pi \end{cases}$$

$$(ii) f(t) = k \frac{t}{T} \text{ for } 0 < t < T \text{ and } f(t+T) = f(t).$$

(b) Using Laplace transform, evaluate $\int_0^\infty \frac{e^{-at} - e^{-bt}}{t} dt.$

$$6 + 4 = 10$$

7. (a) Find $L^{-1} \left\{ \frac{4}{2s-3} - \left(\frac{3+4s}{9s^2-16} \right) + \frac{8-6s}{16s^2+9} \right\}.$

(b) Using Convolution theorem, find the inverse Laplace transform of $\frac{1}{(s-1)^2(s-2)^3}.$

$$5 + 5 = 10$$

8. (a) Solve the differential equation using Laplace transform:

$$(D^3 - 2D^2 + 5D)y = 0 \text{ with } y(0) = 0, y'(0) = 0, y''(0) = 1.$$

- (b) $x_1 = 1, x_2 = 3, x_3 = 2$ is a feasible solution of the L.P.P.:

$$\text{Maximize } Z = x_1 + 2x_2 + 4x_3$$

$$\text{Subject to } 5x_1 + 4x_2 - x_3 = 15,$$

$$2x_1 + 7x_2 - 4x_3 = 15,$$

$$x_1, x_2, x_3 \geq 0.$$

Find the basic feasible solutions from it.

$$5 + 5 = 10$$

9. (a) Define convex set and show that the point set given by

$$X = \{(x_1, x_2) : 4x_1^2 + 9x_2^2 \leq 36\}$$

is a convex set.

- (b) Solve the following LPP by simplex method:

$$\text{Maximize } Z = x_1 + x_2 + 3x_3$$

$$\text{Subject to } 3x_1 + 2x_2 + x_3 \leq 3,$$

$$2x_1 + x_2 + 2x_3 \leq 2,$$

$$x_1, x_2, x_3 \geq 0.$$

$$4 + 6 = 10$$

10. Solve the following LPP by Charnes' Big-M method:

$$\text{Maximize } Z = x_1 + 2x_2 + 3x_3 - x_4$$

$$\text{Subject to } x_1 + 2x_2 + 3x_3 = 15$$

$$2x_1 + x_2 + 5x_3 = 20$$

$$x_1 + 2x_2 + x_3 + x_4 = 10$$

$$x_1, x_2, x_3, x_4 \geq 0.$$

$$10$$